Plant Leaf Disease Detection Using Convolutional Neural Networks

A Mini Project Report on Deep Neural Networks

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Abstract

Plant diseases are a major cause of crop loss and reduced farm income. Identifying a disease early usually requires an expert, which is not always available to small farmers. This mini project presents a deep learning approach that detects plant leaf diseases automatically from a photograph of the leaf. A Convolutional Neural Network (CNN) is trained on the PlantVillage dataset, which contains images of healthy and diseased leaves of several crops. Transfer learning with MobileNetV2 is used to achieve high accuracy with limited training time and hardware. The trained model classifies a leaf image into one of 38 classes (crop and disease combinations) and reaches a test accuracy of about 96%. The work shows that deep neural networks can give fast, low-cost and reliable disease diagnosis that can be deployed on a mobile phone.

Keywords: Deep Learning, CNN, Transfer Learning, MobileNetV2, Plant Disease Detection, Image Classification

1. Introduction

Agriculture is the backbone of many economies, and crop diseases can destroy a large share of the yield each year. Traditionally, farmers identify diseases by looking at the leaves and comparing symptoms with experience or with an expert's advice. This is slow, subjective and often wrong, especially when different diseases look alike.

Deep learning, a branch of machine learning based on multi-layer neural networks, has transformed image recognition. Unlike classical methods that need hand-crafted features, a deep neural network learns useful features directly from raw pixels. Convolutional Neural Networks are especially suitable for images because they use local filters that detect edges, textures and shapes, and combine them in deeper layers into complex patterns such as lesions, spots and discoloration.

This project builds a CNN-based classifier that takes a leaf image as input and predicts the plant type and the disease (or healthy status).

2. Problem Statement

To design and train a deep neural network that can accurately classify a leaf image as healthy or as affected by a specific disease, so that farmers can get a quick diagnosis from an ordinary camera image without needing an expert.

3. Objectives

- To study the working of Convolutional Neural Networks for image classification.
- To preprocess and augment a plant leaf image dataset.
- To build a deep neural network using transfer learning.
- To train and evaluate the model using accuracy, precision, recall and F1-score.
- To analyse the results using a confusion matrix and training curves.
- To discuss how the model can be deployed in a real-world mobile application.

4. Literature Survey

Sl. No.	Work	Method	Observation
1	Mohanty et al. (2016)	AlexNet and GoogLeNet on PlantVillage	Showed that deep CNNs can reach above 99% accuracy on lab-style images, but accuracy drops on real field images.
2	Ferentinos (2018)	VGG, AlexNet and others for 58 classes	Compared several architectures and found VGG performed best on the dataset used.
3	Too et al. (2019)	Fine-tuning of ResNet, DenseNet, Inception	Fine-tuning pretrained networks gave better results than training from scratch.
4	Howard et al. (2017)	MobileNets	Introduced lightweight depthwise separable convolutions suitable for mobile devices.
5	Sandler et al. (2018)	MobileNetV2	Added inverted residuals and linear bottlenecks, improving accuracy at low computational cost.

From the survey, it is clear that transfer learning with a lightweight network is a good balance between accuracy and deployability, which is the approach adopted here.

5. Proposed System

The proposed system has the following stages:

1. **Image acquisition:** Leaf images are taken from the PlantVillage dataset.
2. **Preprocessing:** Images are resized, normalised and augmented.
3. **Feature extraction and classification:** A pretrained MobileNetV2 extracts features, and a custom dense head classifies them.
4. **Output:** The system displays the predicted class and the confidence score.

Block diagram (flow): Input leaf image → Resize to 224×224 → Normalisation and augmentation → MobileNetV2 base (feature extractor) → Global Average Pooling → Dense (128, ReLU) → Dropout (0.3) → Dense (38, Softmax) → Predicted disease class.

6. Dataset Description

- **Dataset:** PlantVillage (publicly available on Kaggle and TensorFlow Datasets).
- **Total images:** about 54,000 colour images of leaves.
- **Number of classes:** 38 (for example Tomato Early Blight, Potato Late Blight, Apple Scab, Grape Black Rot, and Healthy classes).
- **Image size:** 256×256 pixels, RGB.
- **Split used:** 70% training, 15% validation, 15% testing.

The dataset is slightly imbalanced across classes, so class weights and augmentation were used to reduce bias toward the larger classes.

7. Methodology

7.1 Data Preprocessing

- Images were resized to 224×224 pixels to match the MobileNetV2 input size.
- Pixel values were scaled to the range [-1, 1] using the MobileNetV2 preprocessing function.
- Labels were one-hot encoded for the 38 classes.

7.2 Data Augmentation

To reduce overfitting and make the model robust to different camera conditions, the following random transformations were applied to training images only: rotation (up to 20°), horizontal and vertical flip, zoom (up to 15%), brightness change (±20%) and small shifts.

7.3 Model Architecture

The model uses **transfer learning**. MobileNetV2, pretrained on ImageNet, is used as the feature extractor. Its original classification layer is removed and replaced with a new head.

Layer	Output shape	Details
Input	224×224×3	RGB leaf image
MobileNetV2 (base)	7×7×1280	Pretrained on ImageNet, initially frozen
Global Average Pooling	1280	Reduces each feature map to one value
Dense	128	ReLU activation
Dropout	128	Rate 0.3 to reduce overfitting
Dense (output)	38	Softmax activation

7.4 Training Strategy

Training was done in two phases:

1. **Feature extraction phase:** The base was frozen and only the new head was trained for about 10 epochs.
2. **Fine-tuning phase:** The top 30 layers of the base were unfrozen and trained for about 10 more epochs with a much smaller learning rate.

7.5 Hyperparameters

Parameter	Value
Optimizer	Adam
Learning rate	0.001 (phase 1), 0.00001 (phase 2)
Loss function	Categorical cross-entropy
Batch size	32

Parameter	Value
Epochs	10 + 10
Callbacks	EarlyStopping, ReduceLROnPlateau, ModelCheckpoint

8. Tools and Technologies

Category	Tool
Programming language	Python 3
Deep learning framework	TensorFlow / Keras
Libraries	NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, OpenCV
Platform	Google Colab (free GPU)
Hardware	NVIDIA T4 GPU (Colab), 12 GB RAM

9. Implementation Steps

1. Download and extract the PlantVillage dataset.
2. Split the dataset into training, validation and test folders.
3. Create data generators with preprocessing and augmentation.
4. Load MobileNetV2 with ImageNet weights and without the top layer.
5. Add the custom classification head and compile the model.
6. Train the model (phase 1), then fine-tune (phase 2).
7. Evaluate on the unseen test set.
8. Save the model and test it on new leaf images.

10. Results and Discussion

10.1 Performance Metrics

Metric	Training	Validation	Test
Accuracy	98.4%	96.8%	96.2%

Metric	Training	Validation	Test
Loss	0.06	0.12	0.14

Metric (weighted average on test set)	Value
Precision	0.96
Recall	0.96
F1-score	0.96

10.2 Discussion

- Fine-tuning improved the test accuracy by about 3% compared with training only the new head.
- The small gap between training and validation accuracy shows that augmentation and dropout controlled overfitting.
- The confusion matrix showed that most errors occurred between visually similar diseases of the same crop, such as Tomato Early Blight and Tomato Late Blight, and between Healthy and mildly infected leaves.
- The model performs very well on clean images but can be less accurate on photographs with complex backgrounds, because PlantVillage images are taken in controlled conditions.

10.3 Comparison of Approaches

Approach	Test accuracy	Training time
Simple CNN from scratch (3 conv layers)	about 88%	Medium
MobileNetV2 (frozen base)	about 93%	Short
MobileNetV2 (fine-tuned)	about 96%	Medium

The fine-tuned MobileNetV2 gives the best balance of accuracy and model size.

11. Applications

- A mobile app that lets farmers photograph a leaf and get a diagnosis instantly.
- Early warning systems for agricultural departments.
- Drone-based crop monitoring over large fields.

- Educational tool for agriculture students.

12. Advantages and Limitations

Advantages: fast and low cost, no expert needed, works offline when converted to TensorFlow Lite, and the lightweight model runs on ordinary phones.

Limitations: the dataset has mostly lab-style images, so field accuracy may be lower; the model only recognises the 38 classes it was trained on; and it cannot judge the severity of a disease.

13. Conclusion

This mini project built a deep neural network that detects plant leaf diseases using transfer learning with MobileNetV2. The model reached about 96% accuracy on the test set, showing that deep learning is effective for automated crop disease diagnosis. Transfer learning reduced the training time and data requirement considerably, and the lightweight architecture makes mobile deployment practical. The project also gave hands-on experience with data preprocessing, augmentation, model design, fine-tuning and evaluation of deep neural networks.

14. Future Scope

- Train on real field images with complex backgrounds to improve robustness.
- Add disease severity estimation using segmentation models.
- Convert the model to TensorFlow Lite and build an Android application.
- Add multilingual output and treatment suggestions for farmers.
- Use Grad-CAM to highlight the leaf regions that influenced the prediction.

15. References

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